Class 11 Chemistry

NCERT syllabus 2026-27 par based

# Chemophile

**Part 1 (for 11th Class)**

**FREE SAMPLE**
har chapter se ek topic

## Har chapter ki ek jhalak

NCERT Class 11 Chemistry ke saare 9 chapters, easy Hinglish mein, har page par visuals ke saath




1 Some Basic Concepts
2 Structure of Atom
3 Classification of Elements
4 Chemical Bonding
5 Thermodynamics
6 Equilibrium
7 Redox Reactions
8 Organic Chemistry: Basics
9 Hydrocarbons

**651** colourful pages
**499** figures aur diagrams
**333** solved examples
**1010+** practice questions
**9** chapter tests (40 marks)


Adeel Sir

**Mohd Adeel Khan**
MSc BEd (JMI) | Ph. **9953821875**


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# Chemophile kya hai?

**Chemophile Part 1 (for 11th Class)** NCERT Class 11 Chemistry ke saare **9 chapters** ki colourful booklets hain. Explanations **modern Hinglish** mein, technical terms aur exam answers **English** mein. Har topic ek story ki tarah chalta hai: Adeel Sir ka example, visual, concept, exam wala answer, aur end mein practice.

| 9 | 651 | 499 | 333 | 1010+ |
|---|---|---|---|---|
| chapters | colourful pages | figures aur diagrams | solved examples | practice questions |

## Har page par kya milega

| | |
|---|---|
| *Adeel Sir ka example* | Har concept ek everyday life example se start hota hai. |
| ⚠ Galti mat karna | Common mistakes jo sabse zyada marks cut karwati hain. |
| Adeel Sir ka solved example | Step-by-step solved question. |
| Ruko aur socho | Aage padhne se pehle khud socho, phir tick karo. |
| Exam mein aise likho | Board exam ke liye English answer, ruled lines par. |
| ★ JEE/NEET extra | Advanced concepts, competitive exams ke liye. |
| *Adeel Sir ka tip* | Yaad rakhne ki short trick. |
| Ab tumhari baari | Khud solve karo; answers topic ke end mein. |

## Har chapter ke andar

- ✓ Har topic ka pehla page: goals, timeline aur Adeel Sir ka message
- ✓ NCERT ke saare Examples, Intext Questions aur Exercises solved
- ✓ JEE/NEET extra boxes aur common mistakes
- ✓ Har chapter ke end mein revision: fact ya formula sheet, top mistakes
- ✓ Har concept: everyday example, phir visual, phir concept
- ✓ Exam mein aise likho: board ke liye English answers
- ✓ Har topic ke end mein recap, quick check aur answers
- ✓ 40 marks ka chapter test, answer key aur certificate

**Is sample mein kya hai?**

Har chapter se **ek topic** ke do asli pages: topic ka pehla page aur uska pehla lesson page. Phir Chapter 6 ke revision ke do pages (fact sheet aur chapter test ka pehla page). Yeh pages bilkul waise hi hain jaise poori book mein: **651 pages** isi style mein.

*Adeel Sir kehte hain*

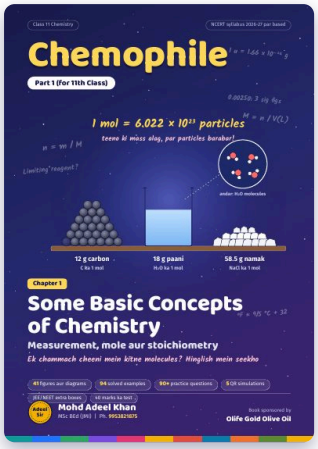
Chemistry ratne ka subject nahi, samajhne ka subject hai. Jab concept dikh jaata hai, to formula khud yaad ho jaata hai.

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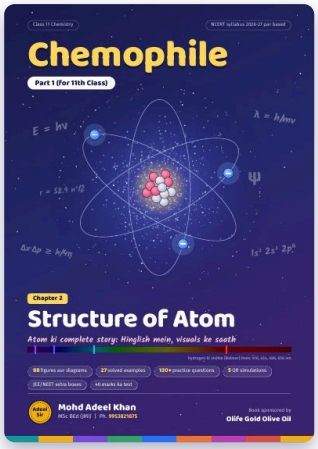

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# Saare 9 chapters ek nazar mein

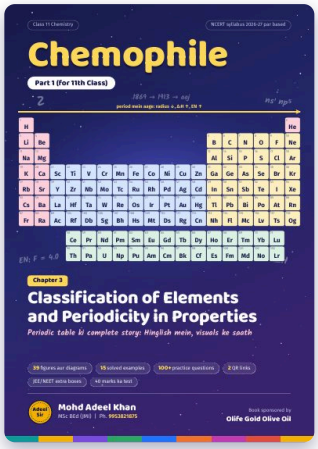
Har chapter ki apni booklet hai. Neeche har booklet ka cover, uske pages aur is sample mein shaamil topic.



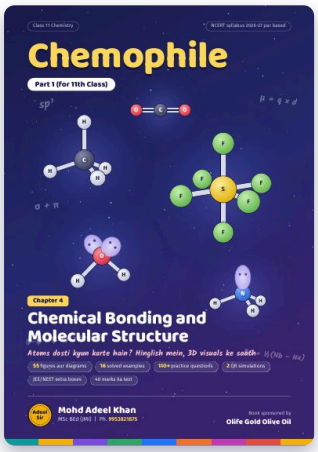

**Chapter 1** Some Basic Concepts
82 pages · Sample: Mole concept



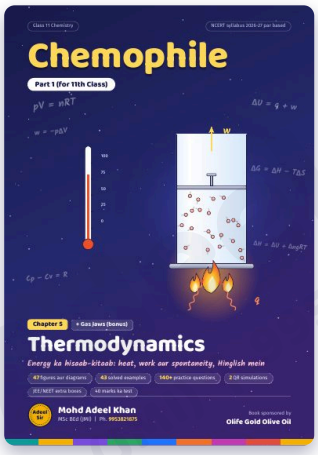

**Chapter 2** Structure of Atom
81 pages · Sample: Bohr model



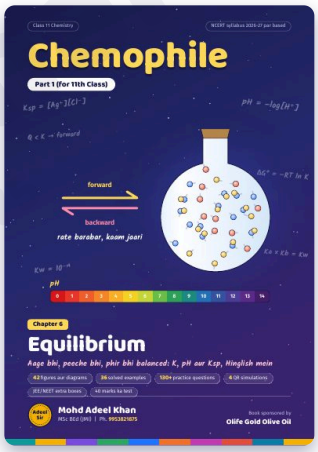

**Chapter 3** Classification of Elements
58 pages · Sample: Ionization enthalpy




**Chapter 4** Chemical Bonding
62 pages · Sample: VSEPR theory: molecules ki shape



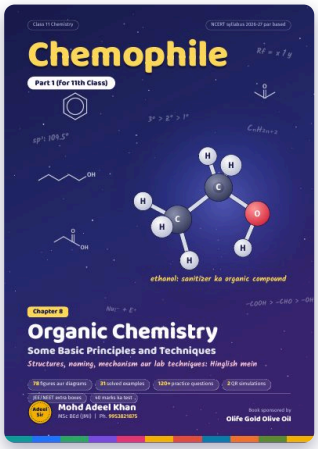

**Chapter 5** Thermodynamics
68 pages · Sample: Reaction enthalpy aur Hess's law



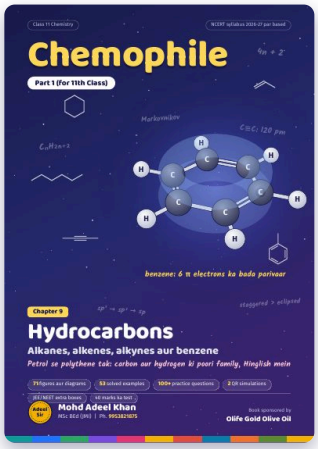

**Chapter 6** Equilibrium
73 pages · Sample: Le Chatelier's principle




**Chapter 7** Redox Reactions
52 pages · Sample: Oxidation number




**Chapter 8** Organic Chemistry: Basics
88 pages · Sample: Electronic effects aur reaction types




**Chapter 9** Hydrocarbons
87 pages · Sample: Benzene: structure aur aromaticity

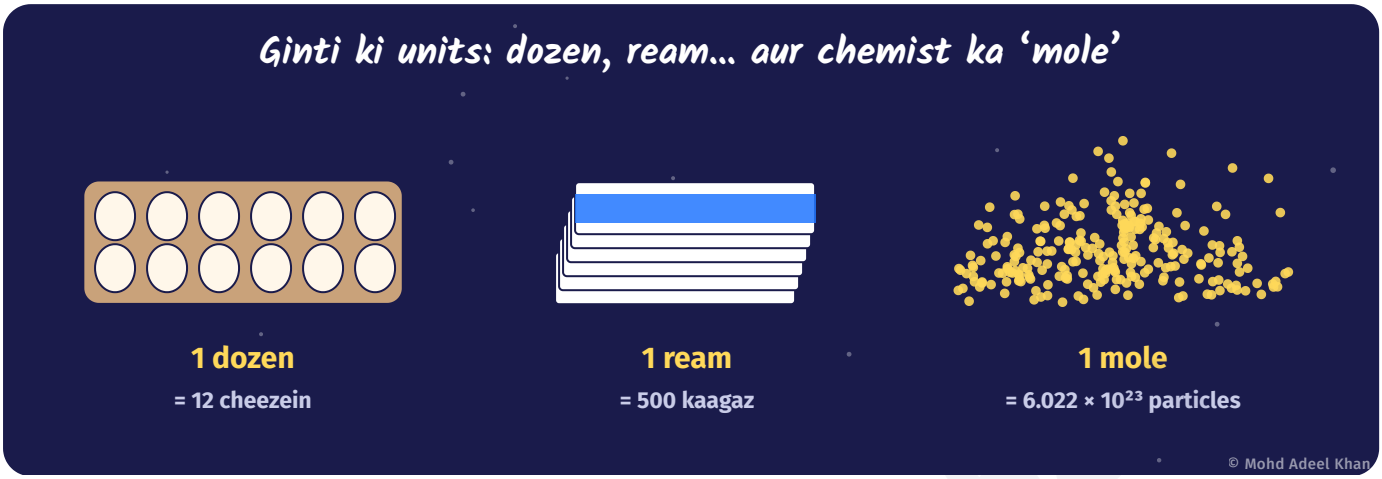
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Topic 6

# Mole concept

*Chemist ka dozen: 6.022 × 10²³*




Atoms aur molecules itne chhote hain ki unhe ek-ek karke gin nahi sakte. Isliye chemistry ne ek bahut badi ginti wali unit banayi: **mole**. Ek mole mein **6.022 × 10²³** particles hote hain (**Avogadro constant**), aur ek mole ka mass gram mein **molar mass** kehlata hai. Is topic mein mole ki definition, molar mass, aur **mass ↔ moles ↔ particles ↔ volume** ke conversions seekhenge.

## Is topic mein


| | |
|---|---|
| Mole aur Avogadro constant | 38 |
| Molar mass aur mole ka bridge | 39 |
| Mole ke numericals + Exercise 1.30 | 40 |
| Exercises 1.10, 1.28, 1.33 | 41 |
| Practice: mole bridge | 42 |
| Ek nazar mein + quick check | 43 |


## Padhne ke baad tick karo

- ☐ Mole aur Avogadro constant define kar sakta hoon
- ☐ Molar mass ko atomic/molecular mass se jod sakta hoon
- ☐ Mass, moles, particles aur gas volume ke beech conversion kar sakta hoon
- ☐ NCERT Exercises 1.10, 1.28, 1.30, 1.33 solve kar sakta hoon

Adeel Sir

***Adeel Sir kehte hain***

Mole se darna nahi: yeh bas ek **bahut bada dozen** hai. Dozen mein 12, mole mein 6.022 × 10²³.

## Kab kya hua

**1811** — Avogadro ki hypothesis

**1909** — Perrin: $N_A$ naapa

**2019** — $N_A$ exact fix

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Section 1.8

Samajh aaya? Rate karo ☆ ☆ ☆

# Mole aur Avogadro constant

### Adeel Sir ka example

Adeel Sir mandi se **do dozen kele** laaye: matlab 24 kele. Kele gin sakte hain, par ek chammach paani ke molecules nahi. Isliye chemist **mole** naam ki ginti use karte hain: '2 mol paani' matlab 2 × 6.022 × $10^{23}$ molecules.

**Mole: amount of substance ki SI unit (mol)**

**Ek mole mein exactly 6.02214076 × $10^{23}$ elementary entities** (atoms, molecules, ions ya koi bhi particles) hote hain. Yahi number **Avogadro constant ($N_A$)** hai.

Purani definition bhi yahi number deti hai: **exactly 12 g $^{12}$C** mein jitne atoms. Ek $^{12}$C atom ka mass 1.992648 × $10^{-23}$ g hai, to 12 g / 1.992648 × $10^{-23}$ g = 6.0221 × $10^{23}$ atoms.

**$N_A = 6.022 \times 10^{23}\ mol^{-1}$ | particles = n × $N_A$**

**Kis cheez ka mole?**

- 1 mol H atoms = 6.022 × $10^{23}$ H atoms
- 1 mol $H_2O$ = 6.022 × $10^{23}$ molecules
- 1 mol NaCl = 6.022 × $10^{23}$ formula units ($Na^+$ bhi utne, $Cl^-$ bhi utne)

★ JEE/NEET extra

**1 mol $H_2O$ mein atoms kitne?** Har molecule mein 3 atoms, to 3 × 6.022 × $10^{23}$ = 1.807 × $10^{24}$ atoms (2 mol H atoms + 1 mol O atoms). Hamesha poochho: **kis particle ki** ginti chahiye?

**Adeel Sir ka solved example 6.1**

**Adeel Sir's question: How many N atoms and H atoms are present in 0.50 mol $NH_3$?**

**N atoms** 1 molecule mein 1 N: 0.50 mol N = 0.50 × 6.022 × $10^{23}$ = 3.011 × $10^{23}$

**H atoms** 1 molecule mein 3 H: 1.50 mol H = 1.50 × 6.022 × $10^{23}$ = 9.033 × $10^{23}$

**N: 3.011 × $10^{23}$; H: 9.033 × $10^{23}$**

⚠ **Galti mat karna.** '1 mol oxygen' adhoora hai: 1 mol O **atoms** (16.00 g) ya 1 mol $O_2$ **molecules** (32.00 g)? Particle ka naam hamesha likho.

**Ruko aur socho**

How many moles of oxygen atoms are present in 2 mol $CO_2$?

☐ 2 mol ☐ 4 mol ☐ 6 mol

**Exam mein aise likho**

**Mole and Avogadro constant**

The mole is the SI unit of the amount of substance. One mole contains exactly 6.02214076 × $10^{23}$ elementary entities. This number is the Avogadro constant ($N_A$). Equivalently, one mole is the amount of a substance that contains as many particles as there are atoms in exactly 12 g of the carbon-12 isotope: 12 g / 1.992648 × $10^{-23}$ g = 6.0221 × $10^{23}$ atoms.



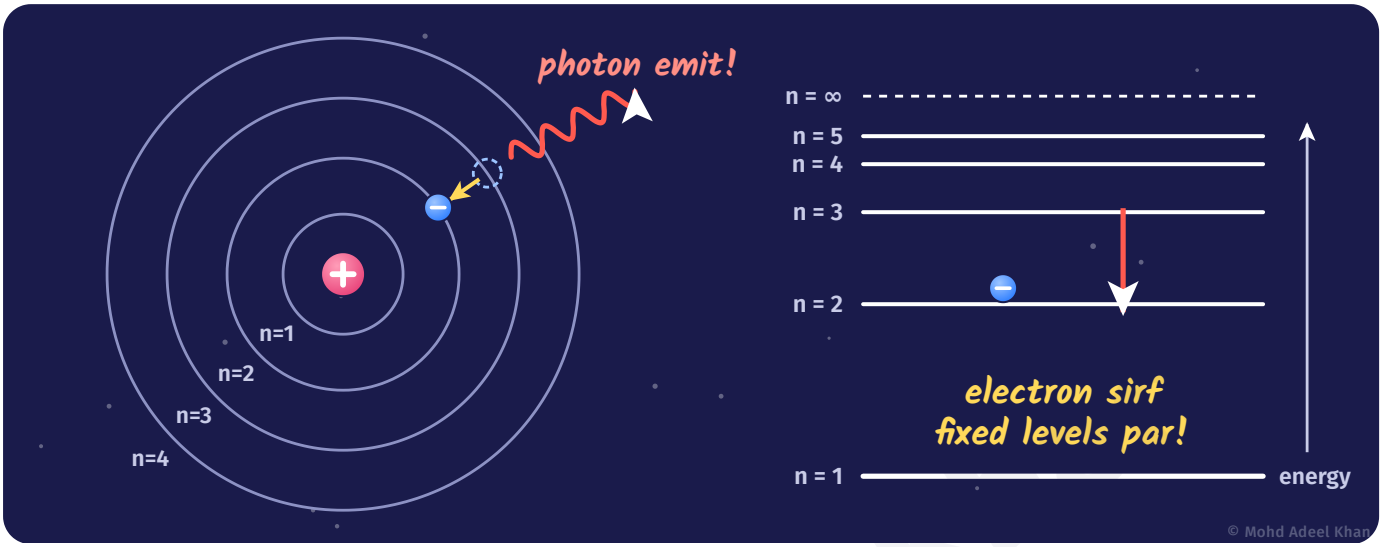
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Topic 5

# Bohr model

*Electron ke fixed energy levels*




Rutherford model ke hisaab se electron ko nucleus mein gir jaana chahiye tha. Aur hydrogen sirf kuch specific lines hi kyun deta hai, iska bhi explanation nahi tha. 1913 mein Niels Bohr ne Planck ka quantum idea atom par apply kiya: **electron ki energy bhi quantised hai**. Dono questions ek saath solve ho gaye.

## Is topic mein


| | |
|---|---|
| Bohr ke 4 postulates | 40 |
| Orbit ka radius aur energy | 41 |
| Hydrogen jaise ions: He+, Li2+ | 42 |
| Hydrogen line spectrum ka explanation | 43 |
| Transition ke numericals | 44 |
| Bohr model ki limitations | 45 |
| Ek nazar mein + quick check | 46 |


## Padhne ke baad tick karo

- [ ] Bohr ke 4 postulates likhna
- [ ] $r_n$ aur $E_n$ ke formulas se radius aur energy nikaalna
- [ ] Negative energy ka matlab explain karna
- [ ] Kisi transition ki energy, frequency aur wavelength nikaalna
- [ ] Bohr model ki limitations likhna

Adeel Sir

***Adeel Sir kehte hain***

Bohr model sirf **single-electron** species par apply hota hai: H, $He^{+}$, $Li^{2+}$, $Be^{3+}$. Numerical dekhte hi pehle yahi check karo.

## Kab kya hua

**1900** — Planck: energy quantised hai

**1911** — Rutherford: nuclear model

**1913** — Bohr: hydrogen atom ka model

**1922** — Bohr ko Physics ka Nobel Prize

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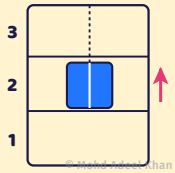
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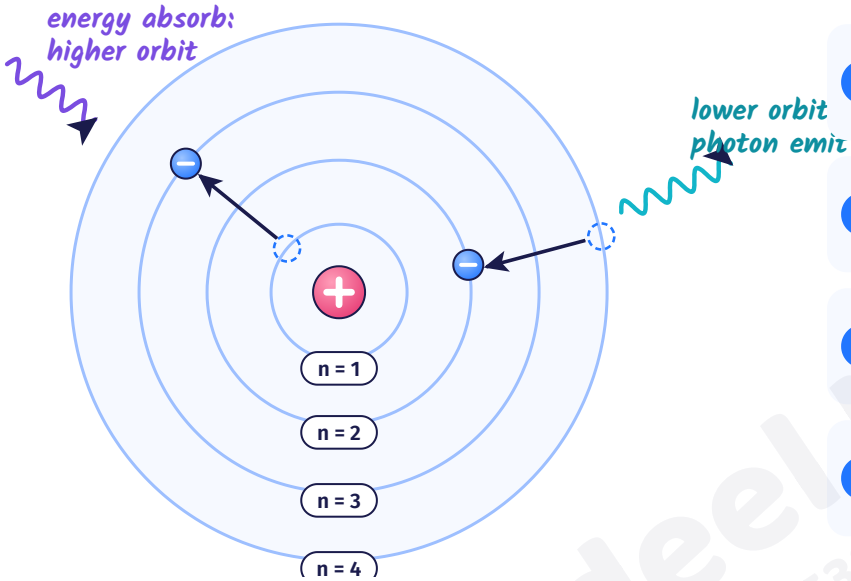
*Samajh aaya? Rate karo* ☆ ☆ ☆

# Bohr ke 4 postulates

### *Adeel Sir ka example*

Adeel Sir ki building ki lift sirf floors par rukti hai: first, second, third floor. Do floors ke beech kabhi nahi. Bohr ne kaha electron bhi aisa hi hai: woh sirf kuch **fixed orbits** mein reh sakta hai, aur orbit change karne ke liye use energy absorb ya emit karni padti hai.




**1 Fixed orbits**
Electron sirf kuch fixed circular orbits mein revolve karta hai.

**2 Orbit mein energy fixed**
Jab tak electron usi orbit mein hai, uski energy constant rehti hai.

**3 Transition = photon**
Energy difference photon ke form mein absorb/emit hota hai: $\Delta E = h\nu$

**4 Angular momentum**
$m_e vr = n \times h/2\pi$
(n = 1, 2, 3 ...)

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**Exam mein aise likho**

**Bohr's model for hydrogen atom: postulates**

i. The electron moves around the nucleus in circular paths of fixed radius and energy, called orbits, stationary states or allowed energy states.

ii. The energy of an electron in an orbit does not change with time. Energy is absorbed when it moves to a higher state and emitted when it moves to a lower state.

iii. The frequency of radiation absorbed or emitted is $\nu = \Delta E/h = (E_2 - E_1)/h$ (Bohr's frequency rule).

iv. The angular momentum of the electron is quantised: $m_e vr = n(h/2\pi)$, n = 1, 2, 3 ...

**Angular momentum kya hai?**

Straight line motion mein momentum = m × v. Circular motion ke liye angular momentum = **$m_e vr$**. Bohr ke hisaab se yeh sirf $h/2\pi$ ke whole-number multiples ho sakta hai.

**⚠ Galti mat karna**

- 'Stationary state' ka matlab yeh nahi ki electron ruka hua hai. Matlab: us orbit mein uski **energy fixed** hai.
- Do orbits ke beech mein electron kabhi nahi milta. Transition direct hota hai.

**Ruko aur socho**

An electron jumps from n = 3 to n = 1. Is energy absorbed or emitted?

☐ absorbed ☐ emitted (a photon is released)

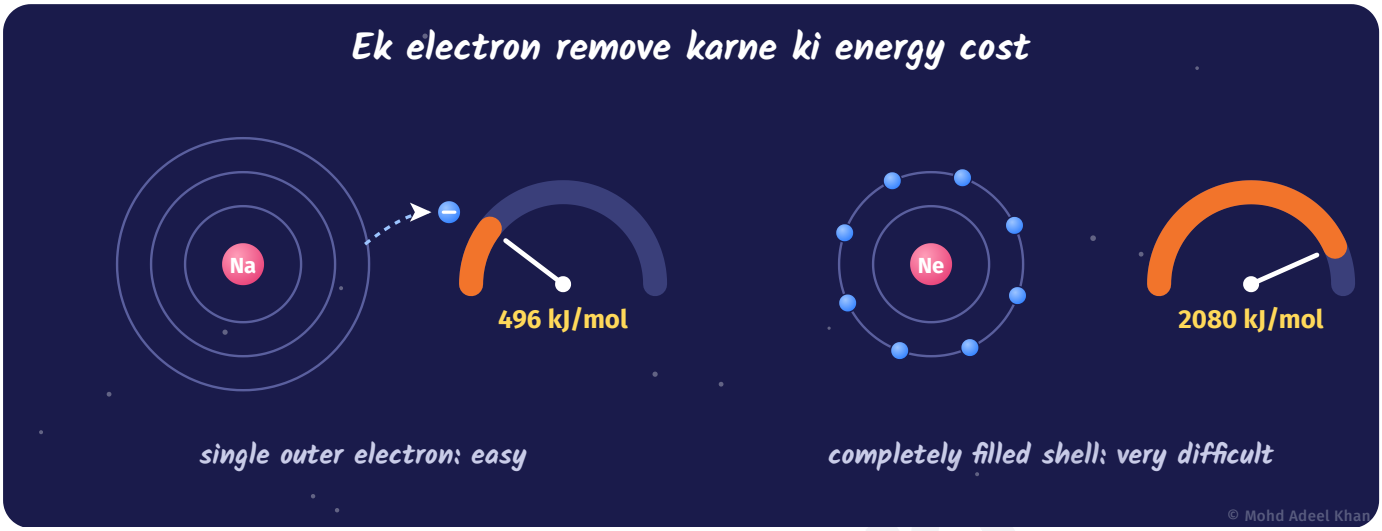

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Topic 6

# Ionization enthalpy

*Electron remove karne mein kitni energy?*




Atom se ek electron remove karne ke liye energy lagti hai. Kuch atoms easily electron lose karte hain (jaise Na), kuch bahut mushkil se (jaise Ne). Yeh energy batati hai ki element kitna **metallic** aur kitna **reactive** hoga. Is topic mein iska trend aur do famous irregularities samjhenge.

## Is topic mein


| | |
|---|---|
| Definition aur successive ionization | 33 |
| ΔiH ka graph: peaks aur minima | 34 |
| Period aur group mein trend; affecting factors | 35 |
| Be > B aur N > O: do irregularities | 36 |
| Ek nazar mein + quick check | 37 |


## Padhne ke baad tick karo

- ☐ Ionization enthalpy ki definition aur equation likh sakta hoon
- ☐ Bata sakta hoon ki $\Delta_iH_2$ hamesha $\Delta_iH_1$ se zyada kyun
- ☐ Period aur group ka trend reason ke saath explain kar sakta hoon
- ☐ Be > B aur N > O ka reason bata sakta hoon

Adeel Sir

***Adeel Sir kehte hain***

Electron nucleus ke jitna **paas** aur nuclear charge jitna **zyada**, electron remove karna utna **difficult**. Poora topic isi ek line par based hai.

Adeel Sir

***Adeel Sir ka tip***

Ionization enthalpy **low** = electron easily remove hota hai = element **metallic**. Isliye alkali metals sabse reactive metals hain.


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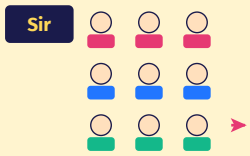
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*Samajh aaya? Rate karo* ☆ ☆ ☆

# Ionization enthalpy kya hai?

### *Adeel Sir ka example*

Adeel Sir ki class mein back bench ka student chupke se bahar nikal jaata hai, par Sir ke bilkul saamne baitha student nahi nikal paata. Atom mein bhi aisa hi hai: nucleus se **door** wala electron easily remove hota hai, paas wala bahut mushkil se.




Ground state ke ek **isolated gaseous atom** se sabse loosely bound electron ko remove karne mein jo energy lagti hai, use **ionization enthalpy** ($\Delta_iH$) kehte hain.

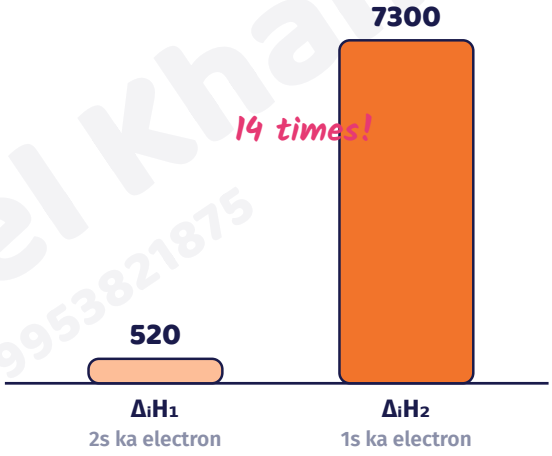
$$X(g) + \text{energy} \rightarrow X^+(g) + e^-$$

$\Delta_iH$ hamesha **positive** (energy supply karni padti hai). Unit: kJ $mol^{-1}$

### Second electron aur bhi difficult

$X^+(g) \rightarrow X^{2+}(g) + e^-$ ko **second ionization enthalpy** ($\Delta_iH_2$) kehte hain. Positive ion se electron remove karna aur difficult hai, kyunki ab kam electrons par same nuclear charge act karta hai. Isliye hamesha **$\Delta_iH_2 > \Delta_iH_1$.**

Lithium mein yeh difference bahut bada hai: second electron **1s** ki completely filled shell se remove karna padta hai.




**Ab tumhari baari** *answers topic ke end mein*

1. For Mg, $\Delta_iH_1$ = 737 and $\Delta_iH_2$ = 1451 kJ $mol^{-1}$. Why is the second value higher?

   Answer

2. Will $\Delta_iH_2$ of Na be very large or only slightly larger? Why?

   Answer

**⚠ Galti mat karna**

- Definition mein **isolated gaseous atom** aur **ground state** zaroor likho.
- Ionization enthalpy hamesha **positive** hoti hai. Negative likha toh galat.

**Exam mein aise likho**

**Ionization enthalpy**

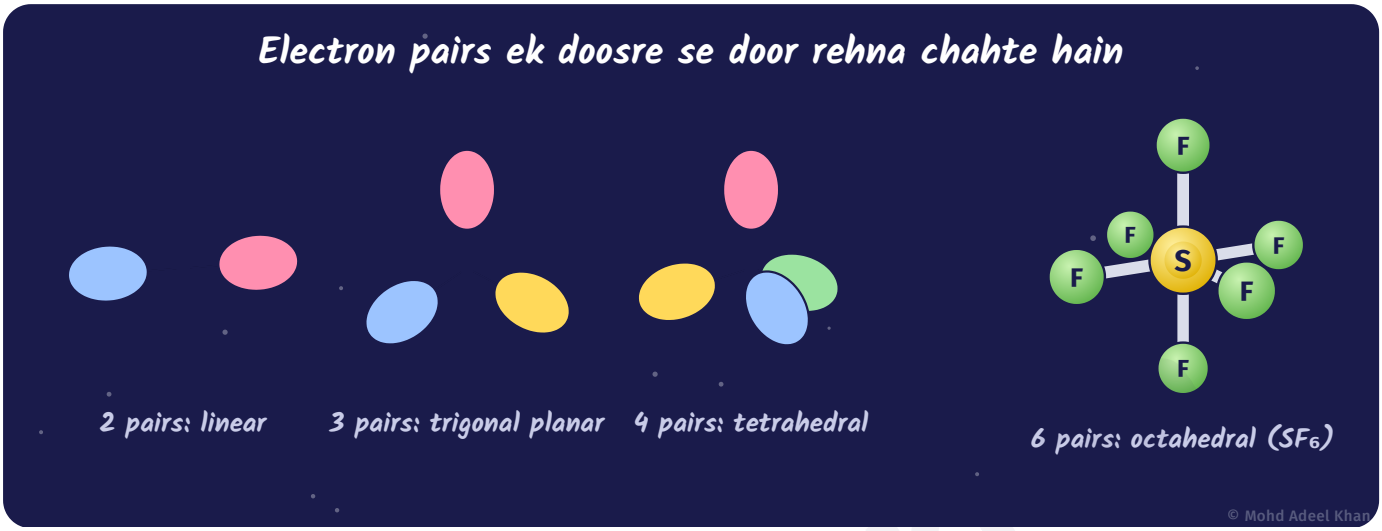
- Ionization enthalpy is the energy required to remove an electron from an isolated gaseous atom (X) in its ground state: $X(g) \rightarrow X^+(g) + e^-$. It is expressed in kJ $mol^{-1}$.
- The second ionization enthalpy is always higher than the first, because it is more difficult to remove an electron from a positively charged ion than from a neutral atom.

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Topic 5

# VSEPR theory: molecules ki shape

*Electron pairs ka 'social distancing'*




Lewis structure batata hai ki kaun kisse juda hai, par molecule **3D mein kaisa dikhta hai**, yeh nahi batata. VSEPR ka simple idea: central atom ke saare electron pairs ek doosre ko **repel** karte hain aur maximum door chale jaate hain.

## Is topic mein


| | |
|---|---|
| VSEPR ke postulates aur repulsion order | 29 |
| Shapes bina lone pair ke ($AB_2$ se $AB_6$) | 30 |
| Shapes jab lone pairs bhi hon | 31 |
| Bond angle distortion aur lone pair ki position | 32 |
| Shape predict karne ke steps + practice | 33 |
| Ek nazar mein + quick check | 34 |


## Padhne ke baad tick karo

- ☐ VSEPR ke postulates likh sakta hoon
- ☐ lp–lp, lp–bp, bp–bp repulsion ka order aur reason bata sakta hoon
- ☐ $AB_2$ se $AB_6$ tak shapes aur bond angles jaanta hoon
- ☐ Lone pairs wale molecules ($NH_3$, $H_2O$, $SF_4$, $ClF_3$, $XeF_4$) ki shape bata sakta hoon
- ☐ Kisi bhi molecule ke liye step-by-step shape predict kar sakta hoon

Adeel Sir

***Adeel Sir kehte hain***

Shape ka naam sirf **atoms ki position** se rakhte hain, par shape decide **saare pairs** (lone pairs bhi) karte hain. Yeh fark samajh liya toh VSEPR easy hai.

## Kab kya hua

**1940**
Sidgwick aur Powell: VSEPR

**1957**
Nyholm aur Gillespie: lone pair effect

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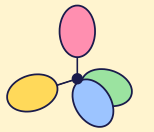
Section 4.4

Samajh aaya? Rate karo ☆ ☆ ☆

# VSEPR theory ka basic idea

### Adeel Sir ka example

Adeel Sir ne birthday par balloons ko beech se ek saath baandha. 2 balloons apne aap **seedhi line** (180°) mein, 3 balloons **triangle** (120°) mein, aur 4 balloons **tetrahedron** (109.5°) mein set ho gaye. Har balloon doosre se maximum door jaana chahta hai. Electron pairs bhi bilkul aise hi behave karte hain!



4 balloons: 109.5°

**VSEPR** = Valence Shell Electron Pair Repulsion. Ise **Sidgwick aur Powell** ne 1940 mein diya, aur **Nyholm aur Gillespie** ne 1957 mein refine kiya.

### Main postulates

1. Molecule ki shape central atom ke **valence shell electron pairs** (bonded + lone) ki sankhya par depend karti hai.
2. Electron pairs negatively charged hain, isliye ek doosre ko **repel** karte hain.
3. Pairs aisi positions lete hain jahan **repulsion minimum** aur distance maximum ho.
4. Valence shell ko ek **sphere** maante hain; pairs uske surface par maximum door rehte hain.
5. **Multiple bond** ko ek single electron pair (super pair) ki tarah count karte hain.
6. Agar molecule ke kai resonance structures hon, toh VSEPR kisi bhi structure par apply hota hai.

## Repulsion ka order

**lp – lp** › › › repulsion sabse zyada

**>**

**lp – bp** › › repulsion beech mein

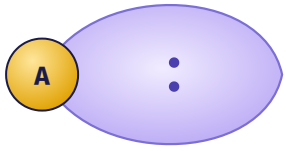
**>**

**bp – bp** › repulsion sabse kam

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**bond pair: do nuclei pull karte hain, slim**




**lone pair: sirf ek nucleus, phaila hua**

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### Lone pair zyada repel kyun?

Lone pair sirf **central atom** par localised hai, jabki bond pair do atoms ke beech share hota hai. Isliye lone pair **zyada space** gherta hai aur zyada repulsion deta hai. Is wajah se ideal shapes distort hote hain aur bond angles badalte hain.

### Ruko aur socho

In VSEPR, how many electron 'groups' are counted around the central C of $CO_2$ (O=C=O)?

☐ 2 ☐ 4



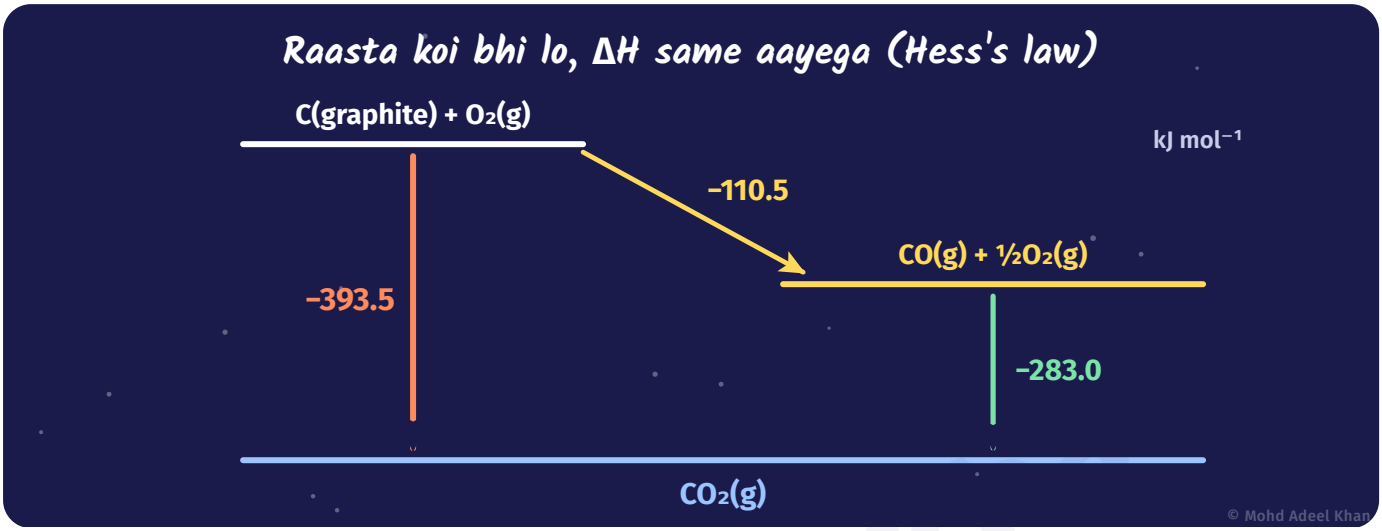
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Topic 6

# Reaction enthalpy aur Hess's law

*Reaction ki heat ka hisaab*




Har reaction ki apni heat hoti hai: LPG jalne par heat nikalti hai, chuna ($CaCO_3$) todne mein heat lagti hai. Is topic mein reaction enthalpy, standard state, enthalpy of formation, thermochemical equations aur **Hess's law** seekhenge, jisse woh ΔH bhi nikal sakte hain jo directly naapna mushkil hai.

## Is topic mein


| | |
|---|---|
| Reaction enthalpy aur standard state | 40 |
| Phase transformations: fusion, vaporisation, sublimation | 41 |
| Standard enthalpy of formation | 42 |
| Thermochemical equations | 43 |
| Hess's law of constant heat summation | 44 |
| Hess's law practice: benzene ka $\Delta_f H$ | 45 |
| Ek nazar mein + quick check | 46 |


## Padhne ke baad tick karo

- [ ] $\Delta_r H$ aur standard state (°) samjha sakta hoon
- [ ] Exothermic aur endothermic ke enthalpy diagrams bana sakta hoon
- [ ] Fusion, vaporisation, sublimation ki enthalpies use kar sakta hoon
- [ ] $\Delta_f H°$ se $\Delta_r H°$ nikaal sakta hoon
- [ ] Thermochemical equations aur Hess's law se numericals kar sakta hoon

Adeel Sir

***Adeel Sir kehte hain***

Hess's law ka simple matlab: ΔH state function hai. Isliye equations ko **jodo, ghatao, ulta karo ya multiply** karo, ΔH ke saath bhi wahi karo.

## Kab kya hua

**1840**
Hess: constant heat summation

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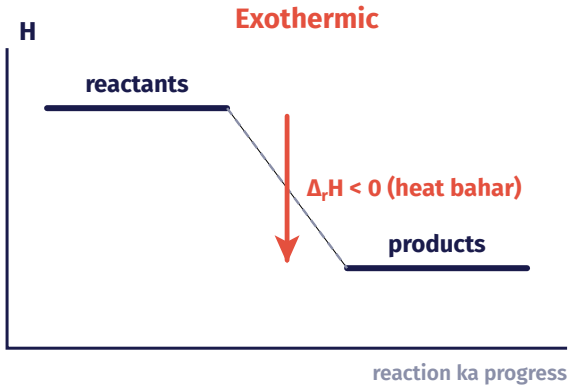

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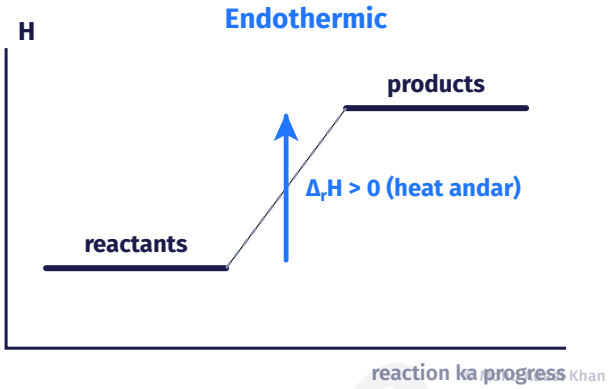
Section 5.4 (a)

*Samajh aaya? Rate karo* ☆ ☆ ☆

# Reaction enthalpy aur standard state

Reaction ke dauraan enthalpy ka change **reaction enthalpy ($\Delta_rH$)** hai: $\Delta_rH$ = (products ki total enthalpy) − (reactants ki total enthalpy) = $\Sigma a_i H_{products} - \Sigma b_i H_{reactants}$ (a, b = balanced equation ke coefficients).




### Standard state kya hai?

Kisi substance ka **standard state** = us temperature par uska **pure form, 1 bar** pressure par. Usually data **298 K** par diya jaata hai.

- 298 K par liquid ethanol ka standard state: pure liquid ethanol, 1 bar.
- 500 K par solid iron ka standard state: pure iron, 1 bar.

### Standard reaction enthalpy ($\Delta_rH°$)

Jab reaction ke saare reactants aur products apne **standard states** mein ho, tab ka reaction enthalpy. Symbol par chhota ° (NCERT mein ⊖, 'plimsoll') = standard conditions.

Reaction enthalpy temperature aur pressure par depend karti hai, isliye standard conditions fix ki gayi hain, taaki sab ka data compare ho sake.

**Ruko aur socho**

What will the enthalpy diagram for burning LPG look like?

- ☐ products lower (exothermic)
- ☐ products higher (endothermic)

**Ab tumhari baari** *answers topic ke end mein*

1. In an endothermic reaction, which has the higher enthalpy, the reactants or the products?

   Answer

2. What is the pressure in the standard state?

   Answer

**Ruko aur socho**

Is photosynthesis ($6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$) exothermic or endothermic?

- ☐ endothermic (it needs the energy of sunlight)
- ☐ exothermic

**Exam mein aise likho**

**Standard enthalpy of reaction**

The standard enthalpy of reaction ($\Delta_rH°$) is the enthalpy change for a reaction when all the participating substances are in their standard states. The standard state of a substance at a specified temperature is its pure form at 1 bar.



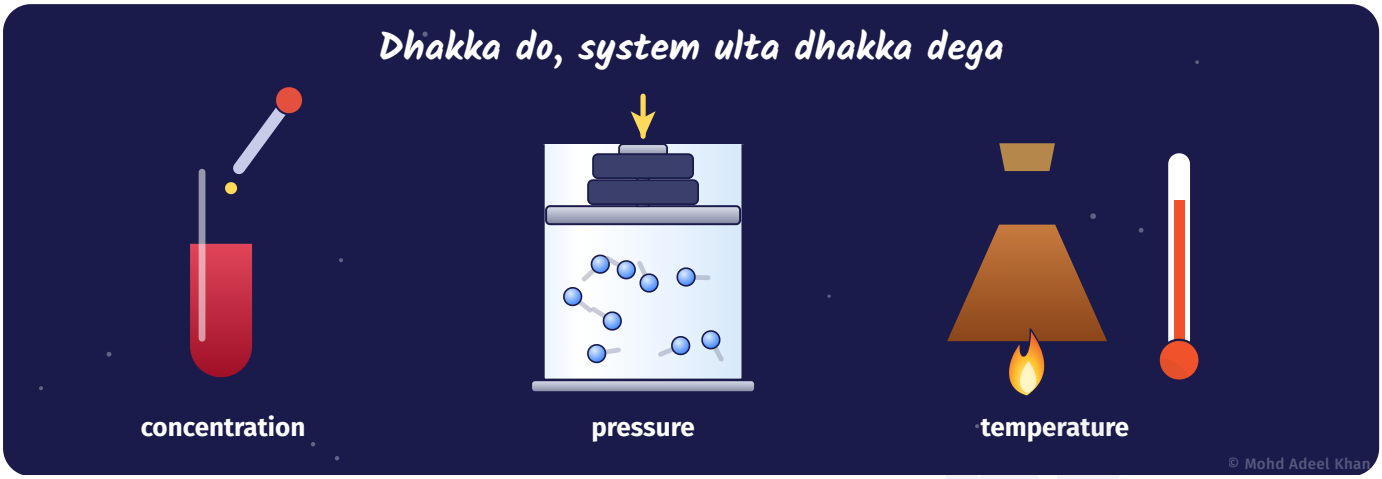
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Topic 5

# Le Chatelier's principle

*Disturb karo, system naya balance dhoondhega*




Equilibrium par baitha system tab tak shaant rehta hai jab tak hum conditions na badlein. Concentration, pressure ya temperature badlo, toh system aise shift hota hai ki us change ka asar **kam** ho jaaye. Yahi **Le Chatelier's principle** hai; industry mein $NH_3$ aur $H_2SO_4$ banane ka poora game isi par chalta hai.

## Is topic mein


| | |
|---|---|
| Principle + concentration ka effect | 31 |
| Pressure aur inert gas ka effect | 32 |
| Temperature: sirf yahi K badalta hai | 33 |
| Catalyst + Haber aur contact process | 34 |
| Le Chatelier master table + practice | 35 |
| Ek nazar mein + quick check | 36 |


## Padhne ke baad tick karo

- [ ] Le Chatelier's principle apne words mein aur English mein likh sakta hoon
- [ ] Concentration change ka effect Q aur K se samjha sakta hoon
- [ ] Pressure aur inert gas ka effect bata sakta hoon
- [ ] Samjha sakta hoon ki sirf temperature hi K ko badalta hai
- [ ] Catalyst ka role aur Haber process ki conditions explain kar sakta hoon

Adeel Sir

***Adeel Sir kehte hain***

System ke saath jo karo, woh **ulta** karta hai: reactant daalo toh use **khapaata** hai, heat do toh heat **absorb** karne wali taraf jaata hai.

## Kab kya hua

**1884**
Le Chatelier ka principle

**1918**
Haber ko Chemistry Nobel

**1931**
Bosch ko Nobel (high-pressure chemistry)

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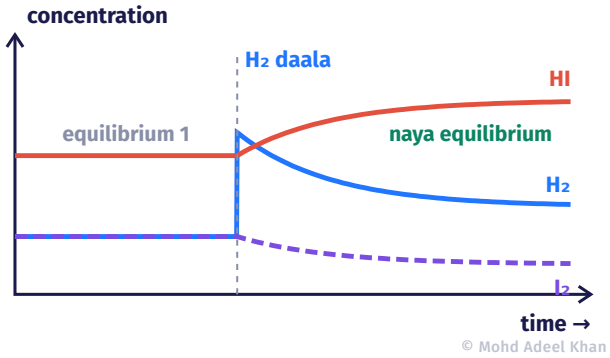
Section 6.8, 6.8.1

*Samajh aaya? Rate karo* ☆ ☆ ☆

# Le Chatelier: concentration badlo

### Le Chatelier's principle

Equilibrium ko decide karne wale kisi bhi factor (concentration, pressure, temperature) mein change karo, toh system **is tarah shift** hota hai ki us change ka effect **kam ya counteract** ho jaaye.



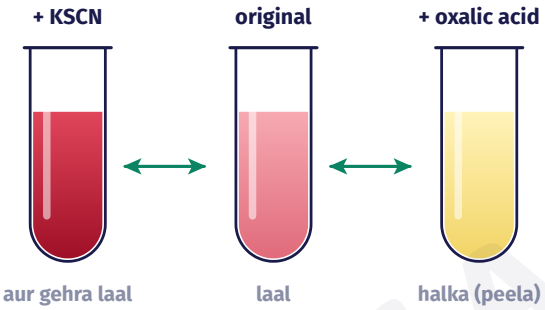

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Fig. 6.8 jaisa: $H_2 + I_2 \rightleftharpoons 2HI$ mein $H_2$ daala

### Kyun shift hota hai?

$H_2$ daalte hi $Q_c = [HI]^2/[H_2][I_2]$ **chhota** ho jaata hai: $Q_c < K_c$. Reaction **forward** chalta hai, extra $H_2$ kuch khapta hai, $I_2$ ghatta hai, HI badhta hai, jab tak $Q_c$ phir $K_c$ nahi ho jaata.

- Reactant **daalo** / product **nikaalo**: forward
- Product **daalo** / reactant **nikaalo**: backward




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### NCERT experiment

$Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons [Fe(SCN)]^{2+}(aq)$
(peela-sa) (colourless) (deep red)

**KSCN** daalo: aur laal. **Oxalic acid** daalo ($Fe^{3+}$ ko $[Fe(C_2O_4)_3]^{3-}$ mein baandh leta hai) ya **$HgCl_2$** ($SCN^-$ ko $[Hg(SCN)_4]^{2-}$ mein): laal colour **halka**.

### Ruko aur socho

Some HI is removed from the $H_2 + I_2 \rightleftharpoons 2HI$ mixture. Which way will the equilibrium shift?

☐ forward ☐ backward ☐ no effect

### Ab tumhari baari *answers topic ke end mein*

1. In the Haber process, $NH_3$ is liquefied and removed continuously. Why does this increase the yield?

Answer

### ★ JEE/NEET extra

**Dilution ka effect:** aqueous equilibrium mein paani daalo toh har concentration ek hi factor se ghatti hai. $Fe^{3+} + SCN^- \rightleftharpoons [Fe(SCN)]^{2+}$ ko double dilute karo: $Q = (c/2)/[(a/2)(b/2)] = 2K > K$, isliye equilibrium **backward** (zyada ions wali taraf) jaata hai aur laal colour halka hota hai.

### Exam mein aise likho

**Le Chatelier's principle: effect of concentration**

Le Chatelier's principle: a change in any of the factors that determine the equilibrium conditions of a system will cause the system to change in such a manner as to reduce or counteract the effect of the change. Adding a reactant or removing a product makes $Q_c < K_c$, so the equilibrium shifts in the forward direction.

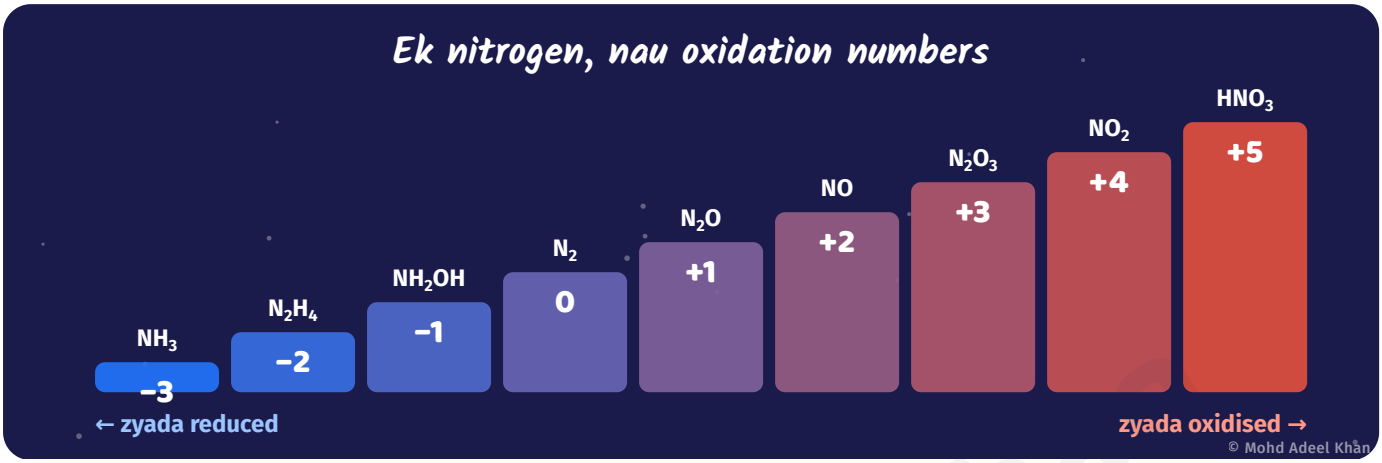
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Topic 3

# Oxidation number

*Electrons ka hisaab-kitaab register*




$H_2 + Cl_2 \rightarrow 2HCl$ mein koi electron poora transfer nahi hota, kyunki HCl covalent hai. Phir bhi yeh redox hai. Aise cases ke liye chemists ne ek simple **hisaab ka tarika** banaya: **oxidation number** (ON). Iske 6 rules seekh lo, phir kisi bhi reaction mein oxidation aur reduction ek nazar mein dikh jaayega.

## Is topic mein


| | |
|---|---|
| Partial transfer aur ON ki definition | 16 |
| ON ke 6 rules | 17 |
| ON calculate karna: solved + practice | 18 |
| Highest ON aur Stock notation (Problem 7.3) | 19 |
| ON se redox pehchaano (Problem 7.4) | 20 |
| Fractional ON: average ka khel | 21 |
| Ek nazar mein + quick check | 22 |


## Padhne ke baad tick karo

- [ ] Oxidation number ki NCERT definition likh sakta hoon
- [ ] ON ke 6 rules use karke kisi compound ya ion mein ON nikaal sakta hoon
- [ ] Stock notation (Roman numerals) mein formulas likh sakta hoon
- [ ] ON badhne/ghatne se oxidation, reduction, oxidant aur reductant pehchaan sakta hoon
- [ ] Fractional ON ka matlab (average) structure se samjha sakta hoon

Adeel Sir

***Adeel Sir kehte hain***

Oxidation number ek **hisaab** hai, real charge nahi. Bond ka electron pair poora **zyada electronegative** atom ke khaate mein daal do; jo charge bachta hai, wahi ON hai.

## Kab kya hua

**1852**
Frankland: valency ka idea

**1919**
Alfred Stock: Roman numeral notation

**1932**
Pauling: electronegativity scale


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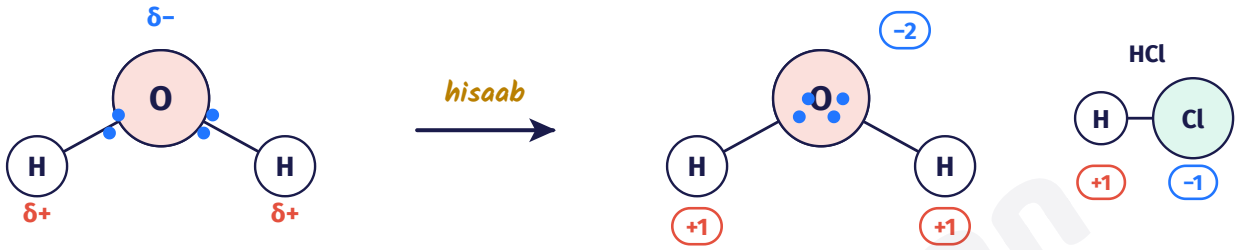
Section 7.3

*Samajh aaya? Rate karo* ☆ ☆ ☆

# Oxidation number kyun chahiye?

### *Adeel Sir ka example*

Adeel Sir ki class mein Riya aur Aman ek samosa share karte hain. Riya zyada 'strong' hai, toh samosa uski taraf zyada khisak jaata hai. Teacher ne register simple rakhne ke liye likh diya: **poora samosa Riya ka**. Chemistry mein bhi yahi kiya jaata hai: bond ka shared electron pair poora **zyada electronegative atom** ke naam likh dete hain. Is register ka naam hai **oxidation number**.




**Asli picture: pair O ki taraf khincha**
(partial charges)

**ON ka hisaab: pair poora O ka**
O: −2, har H: +1 (sum = 0)

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### Covalent reactions bhi redox

$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

$H_2O$ aur HCl covalent hain: electron **poora** transfer nahi hua, bas O aur Cl ki taraf **khisak** gaya (partial transfer). ON ke hisaab se H: 0 → +1 (oxidised), O: 0 → −2 aur Cl: 0 → −1 (reduced).

### Same atoms ka bond

$H_2$, $Cl_2$, $O_2$ mein dono atoms **same** hain: pair barabar baant do. Isliye free element mein har atom ka ON **zero**.

Oxidation number aur **oxidation state** ek hi cheez ke do naam hain.

### Ruko aur socho

In HF, F is more electronegative. What will be the ONs of H and F?

☐ H: +1, F: −1 ☐ H: −1, F: +1 ☐ both zero

Adeel Sir

### *Adeel Sir ka tip*

ON = **atom ke apne valence electrons − hisaab ke baad uske paas electrons.** $H_2O$ mein O ke paas 6 + 2 = 8 hue, toh ON = 6 − 8 = −2.

### Ab tumhari baari *answers topic ke end mein*

**1.** ON of each O in $O_3$ (ozone)?

Answer

**2.** In $NH_3$, N is more electronegative. ONs of N and H?

Answer

### Exam mein aise likho

**Oxidation number**

Oxidation number denotes the oxidation state of an element in a compound, ascertained according to a set of rules formulated on the basis that the electron pair in a chemical bond belongs entirely to the more electronegative element. In $H_2O$, the oxidation number of oxygen is -2 and that of each hydrogen is +1.

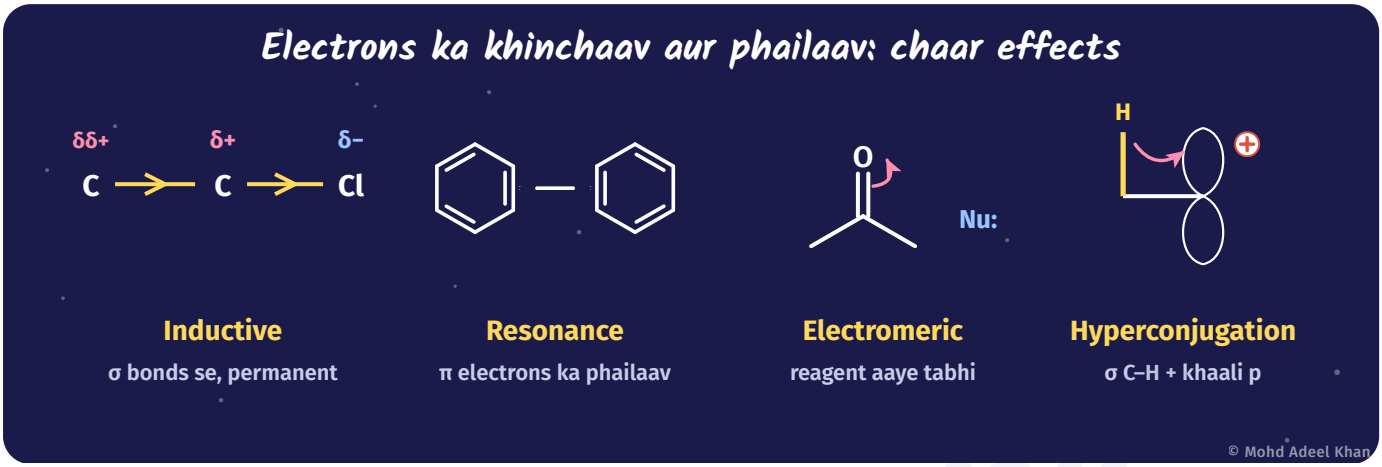

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Topic 8

# Electronic effects aur reaction types

*Electrons kis taraf khisakte hain, aur kyun*




Molecule ke andar electrons hamesha barabar nahi bante: koi atom unhe **kheenchta** hai, koi **deta** hai, aur π electrons **phail** jaate hain. Yeh **electron displacement effects** batate hain ki molecule ka kaun sa hissa electron-rich hai aur kaun sa electron-poor, yaani attack kahan hoga, acid kitna strong hoga, aur intermediate kitna stable. Aakhir mein organic reactions ke chaar types.

## Is topic mein


| | |
|---|---|
| Inductive effect + Problems 8.14, 8.15 | 54 |
| Inductive effect ka use: acidity aur stability | 55 |
| Resonance structures + Problem 8.16 | 56 |
| Resonance structures ki stability + Problems 8.17, 8.18 | 57 |
| Resonance effect: +R aur −R | 58 |
| Electromeric effect: +E aur −E | 59 |
| Hyperconjugation + Problem 8.19 | 60 |
| Organic reactions ke chaar types | 61 |
| Ek nazar mein + quick check | 62 |


## Padhne ke baad tick karo

- [ ] Permanent aur temporary electron displacement effects mein fark bata sakta hoon
- [ ] Inductive effect se acidity aur stability compare kar sakta hoon (Problems 8.14, 8.15)
- [ ] Resonance structures likh kar unki stability bata sakta hoon (Problems 8.16 se 8.18)
- [ ] +R/−R, +E/−E aur hyperconjugation (Problem 8.19) samjha sakta hoon
- [ ] Substitution, addition, elimination aur rearrangement pehchaan sakta hoon

Adeel Sir

***Adeel Sir kehte hain***

Har effect ke liye ek question: **electron density kahan badhi, kahan ghati?** Isi se reactivity aur stability ka answer nikalta hai.

## Kab kya hua

**1916**
Lewis: shared electron pair bond

**1931**
Pauling: resonance theory

**1935**
Baker–Nathan effect (hyperconjugation)

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Section 8.7.4, 8.7.5

*Samajh aaya? Rate karo* ☆ ☆ ☆

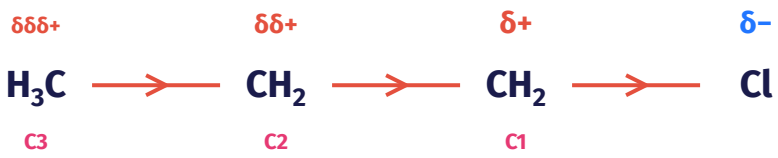
# Inductive effect: σ bonds se khinchaav

**Permanent vs temporary effects**

Molecule ke andar kisi atom/group ki wajah se **permanent polarisation**: **inductive** aur **resonance** effect. Reagent ke paas aane par hi hone wala **temporary** displacement: **electromeric** (polarisability) effect.

***Adeel Sir ka example***

Rassi-kheench (tug of war) socho: Cl sabse taakatwar khiladi hai. Woh C1 ke electrons kheenchta hai, C1 apni kami C2 se poori karta hai, C2 C3 se. Par jitna door, utna kam asar.




Cl electrons kheenchta hai. C1 apni kami C2 se, C2 apni kami C3 se poori karta hai; teen bonds ke baad asar khatam.

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**Inductive effect kya hai?**

Polar σ bond (jaise C–Cl) ki wajah se **paas ke σ bonds bhi polarise** ho jaate hain. Isse **inductive effect** kehte hain. Yeh har agle bond par **tezi se ghatta** hai aur **teen bonds ke baad almost khatam** ho jaata hai. Shift ka arrow δ+ se δ− ki taraf.

**−I groups (electron withdrawing)**

Halogens (F, Cl, Br, I), $-NO_2$, –CN, –COOH, –COOR, –OAr (jaise $-OC_6H_5$)

**+I groups (electron donating)**

Alkyl groups: $-CH_3$, $-CH_2CH_3$ wagairah. Comparison hamesha **H ke relative**.

**Adeel Sir ka solved example 8.1**

**NCERT Problems 8.14, 8.15: (a) Which bond is more polar? $H_3C–H$ / $H_3C–Br$; $H_3C–NH_2$ / $H_3C–OH$; $H_3C–OH$ / $H_3C–SH$ (b) In $CH_3CH_2CH_2Br$, which C–C bond has the least inductive effect?**

(a) Electronegativity ka fark zyada = zyada polar: **C–Br** (Br > H), **C–O** (O > N), **C–O** (O > S)

(b) Br wala C = C1. Bonds jitne door, effect utna kam: C1–C2 se zyada door **C2–C3** bond, isliye wahan sabse kam (C3–H bonds mein to aur bhi kam).

**(a) C–Br, C–O, C–O (b) C2–C3 bond**

**Exam mein aise likho**

**Inductive effect**

When a polar σ bond (e.g. C–Cl) induces polarity in the adjacent σ bonds, it is called the inductive effect. It decreases rapidly with the number of intervening bonds and becomes vanishingly small after three bonds. Electron-withdrawing groups (–I) include halogens, $-NO_2$, –CN, –COOH, –COOR and –OAr; alkyl groups such as $-CH_3$ and $-C_2H_5$ are electron donating (+I).



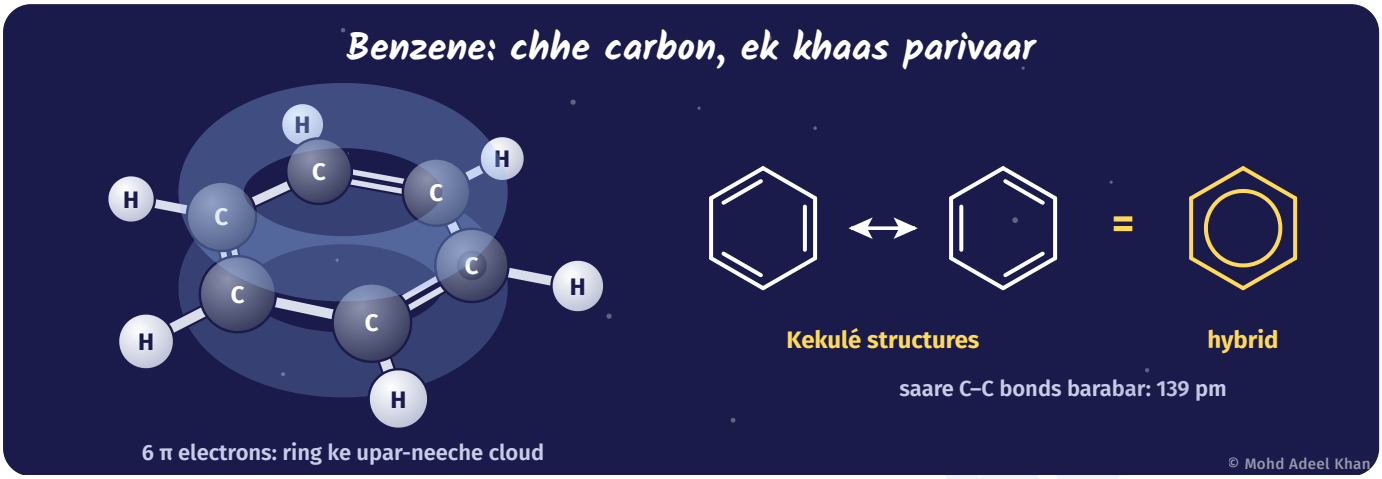
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Topic 9

# Benzene: structure aur aromaticity

*Teen double bonds, phir bhi itna shaant kyun?*




Benzene ring wale hydrocarbons ko **aromatic hydrocarbons** ya **arenes** kehte hain. Shuru mein inki khushboo (aroma) se yeh naam pada, par aaj 'aromatic' ka matlab ek **khaas stability** hai. Is topic mein arenes ke naam aur isomers, benzene ki structure (Kekulé se resonance tak), aromaticity ka **Hückel rule**, aur benzene banane ke tarike seekhenge.

## Is topic mein


| | |
|---|---|
| Arenes: naam aur isomers | **64** |
| Benzene ki structure + Exercise 9.17 | **65** |
| Aromaticity: Hückel rule + Exercises 9.10 to 9.12 | **66** |
| Benzene banana + Problem 9.14, Exercise 9.20 | **67** |
| Physical properties, combustion + Exercise 9.8 | **68** |
| Ek nazar mein + quick check | **69** |


## Padhne ke baad tick karo

- ☐ Arenes ke naam aur o-, m-, p- isomers likh sakta hoon
- ☐ Kekulé structure, uski kami aur resonance hybrid samjha sakta hoon
- ☐ Hückel rule se aromatic compounds pehchaan sakta hoon
- ☐ Benzene banane ke tarike aur Problem 9.14 likh sakta hoon

Adeel Sir

***Adeel Sir kehte hain***

Aromatic ka matlab khushboo nahi: **planar ring + poora delocalised π cloud + (4n + 2) π electrons**. Teeno shartein poori, tabhi aromatic.

## Kab kya hua

**1825**
Faraday: benzene khoja

**1865**
Kekulé: ring structure

**1931**
Hückel: (4n + 2) rule


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Section 9.5, 9.5.1

*Samajh aaya? Rate karo* ☆ ☆ ☆

# Arenes: naam aur isomers

**Benzenoid vs non-benzenoid**

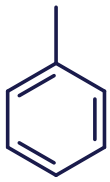
Benzene ring wale aromatic compounds **benzenoid**; jinme benzene ring nahi par phir bhi aromatic hain, woh **non-benzenoid** (Unit 8 mein tropolone). Is unit mein focus benzenoid par.

**Monosubstituted: sirf ek!**

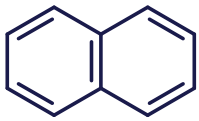
Benzene ke chhe H **equivalent** hain, isliye kisi bhi ek H ko badlo, ek hi product: toluene, chlorobenzene, nitrobenzene ka sirf ek-ek isomer.



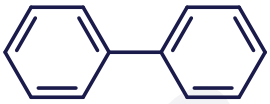
**benzene**
$C_6H_6$



**toluene**
$C_6H_5CH_3$



**naphthalene**
$C_{10}H_8$

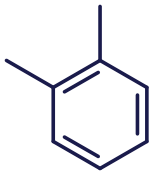


**biphenyl**
$C_6H_5–C_6H_5$

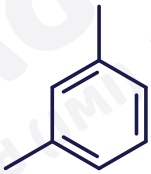
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**Disubstituted: teen position isomers**

Do substituents hon to teen jagah: **ortho (o-)** = 1,2 (ya 1,6); **meta (m-)** = 1,3 (ya 1,5); **para (p-)** = 1,4. Jaise dimethylbenzene (xylene):



**o-xylene**
1,2-dimethylbenzene



**m-xylene**
1,3-dimethylbenzene



**p-xylene**
1,4-dimethylbenzene

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★ **JEE/NEET extra**

**$C_8H_{10}$ ke aromatic isomers: 4**: teen xylenes (o, m, p) aur **ethylbenzene**. Trimethylbenzene ke 3 isomers: 1,2,3-, 1,2,4- aur 1,3,5-.

**Ruko aur socho**

One more Cl is put on chlorobenzene. How many dichlorobenzene isomers are possible?

☐ 1 ☐ 2 ☐ 3

**Exam mein aise likho**

**Isomerism in substituted benzenes**

All six hydrogen atoms of benzene are equivalent, so it forms only one type of monosubstituted product. When two hydrogen atoms are replaced by two similar or different groups, three position isomers are possible: 1,2 or 1,6 (ortho, o-), 1,3 or 1,5 (meta, m-) and 1,4 (para, p-), e.g. o-, m- and p-xylene (dimethylbenzenes).

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Revision

*Samajh aaya? Rate karo* ☆ ☆ ☆

# Fact sheet: poora chapter ek page par

### 1 Physical equilibrium

Closed system; $rate_f = rate_b$; properties constant

Normal m.p./b.p. 1 atm par (paani 273 K / 373 K)

Henry's law: $m \propto p$; T ↑ par gas ki solubility ↓

### 2 Chemical equilibrium, $K_c$

Dynamic: $D_2$ experiment; dono taraf se same equilibrium

$K_c$ sirf temperature se badalta hai

Ulta: 1/K; × n: $K^n$; jodo: $K_1K_2$

### 3 $K_c$, $K_p$, phases

Homogeneous: ek phase; heterogeneous: ek se zyada

$K_p = K_c(RT)^{\Delta n}$; Δn sirf gases

$CaCO_3$: $K_p = p_{CO_2}$ (solids nahi)

### 4 K ka use

$K > 10^3$: products; $K < 10^{-3}$: reactants

Q < K forward; Q > K backward

$\Delta G° < 0 \Leftrightarrow K > 1$; equilibrium par ΔG = 0

### 5 Le Chatelier

Change ka ulta react karta hai

p ↑: kam moles ki taraf; inert gas (V const.): no effect

Haber: approx 200 atm, 700–773 K, Fe catalyst

### 6 Acids aur bases

Arrhenius: paani mein $H^+$ / $OH^-$

Brønsted: $H^+$ donor/acceptor; pair = 1 $H^+$ ka farak

Lewis: $e^-$ pair acceptor ($BF_3$) / donor ($NH_3$)

### 7 pH, $K_a$, $K_b$

$K_w = 10^{-14}$ (298 K); $[H_2O] = 55.55$ M

$\alpha = \sqrt{(K_a/c)}$: dilution par α ↑

HF ≪ HCl < HBr < HI; $K_{a1} > K_{a2} > K_{a3}$

### 8 Common ion, hydrolysis, buffer

Common ion: ionisation dabta hai

Salt ka nature = strong parent; $CH_3COONH_4$: 7.005

Buffer: $pH = pK_a + \log(\text{salt/acid})$; blood 7.4

### 9 $K_{sp}$

Q > $K_{sp}$: precipitate; Q < $K_{sp}$: aur ghulega

Alag formula type: S compare karo, $K_{sp}$ nahi

Common ion: S ↓; weak acid ke salts: kam pH par S ↑

### NCERT ke printed answers: sahi values

6.5: $K_p = 3.33 \times 10^{-4}$ (0.033 nahi); 6.21: $K_b \approx 6.3 \times 10^{-7}$

6.27: $Ni(OH)_2$ ki S = $7.9 \times 10^{-6}$ M

Co complex: $[Co(H_2O)_6]^{2+}$ (pink)

## pH nikaalne ka decision table

| Solution | Formula (298 K) | Example |
|---|---|---|
| Strong acid / strong base | pH = −log C; pOH = −log C | 0.01 M HCl: 2 |
| Weak acid / weak base | $[H^+] = \sqrt{(K_a c)}$; $[OH^-] = \sqrt{(K_b c)}$ | 0.1 M $CH_3COOH$: 2.88 |
| Buffer | $pH = pK_a + \log(\text{salt/acid})$ | 1 : 1 acetate: 4.76 |
| Salt: WA + SB / SA + WB | $7 + \frac{1}{2}(pK_a + \log c)$ / $7 - \frac{1}{2}(pK_b + \log c)$ | 0.1 M $CH_3COONa$: 8.88 |

## Numbers jo exam mein kaam aate hain

**$1.0 \times 10^{-14}$**
$K_w$ at 298 K

**55.55 M**
$[H_2O]$

**0.0831**
R (bar L $mol^{-1}$ $K^{-1}$)

**8.314**
R (J $mol^{-1}$ $K^{-1}$)

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Revision

Samajh aaya? Rate karo ☆ ☆ ☆

# Final test: Part A

*Adeel Sir ke test rules*

Book band karo. Timer lagao: Part A ke liye **15 minutes**. Har correct answer ka **1 mark**. Sirf ek option mark karo.

**Part A:** ____ / 16

**Q1** What becomes equal at equilibrium?
- ◯ concentrations
- ◯ rates
- ◯ masses
- ◯ nothing

**Q2** Henry's law is for what?
- ◯ solid in liquid
- ◯ gas in liquid
- ◯ liquid in gas
- ◯ solid in solid

**Q3** $N_2 + 3H_2 \rightleftharpoons 2NH_3$: unit of $K_c$?
- ◯ mol $L^{-1}$
- ◯ $mol^{-2}\ L^2$
- ◯ $mol^2\ L^{-2}$
- ◯ no unit

**Q4** K for A ⇌ B is 4. K for 2B ⇌ 2A?
- ◯ 1/16
- ◯ 16
- ◯ 1/4
- ◯ −16

**Q5** $K_p$ for $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$?
- ◯ $p_{CO_2}$
- ◯ $1/p_{CO_2}$
- ◯ $p_{CO_2}^2$
- ◯ 1

**Q6** $K_p = K_c$ kab?
- ◯ $\Delta n = 1$
- ◯ $\Delta n = 0$
- ◯ $\Delta n = -1$
- ◯ always

**Q7** $Q_c = 0.5$, $K_c = 10$. Reaction?
- ◯ forward
- ◯ backward
- ◯ equilibrium
- ◯ stopped

**Q8** If $\Delta G° = 0$, K = ?
- ◯ 0
- ◯ 1
- ◯ 10
- ◯ ∞

**Q9** Exothermic reaction: K on raising T?
- ◯ increases
- ◯ decreases
- ◯ same
- ◯ zero

**Q10** On adding an inert gas (constant V)?
- ◯ forward
- ◯ backward
- ◯ no shift
- ◯ K increases

**Q11** Which is a Lewis acid?
- ◯ $NH_3$
- ◯ $H_2O$
- ◯ $BF_3$
- ◯ $OH^-$

**Q12** Conjugate base of $HCO_3^-$?
- ◯ $H_2CO_3$
- ◯ $CO_3^{2-}$
- ◯ $CO_2$
- ◯ $OH^-$

**Q13** pH of 0.001 M NaOH (298 K)?
- ◯ 3
- ◯ 11
- ◯ 10
- ◯ 14

**Q14** $pK_a = 4.76$; $pK_b$ of the conjugate base?
- ◯ 4.76
- ◯ 9.24
- ◯ 14
- ◯ 7

**Q15** Which is a buffer?
- ◯ HCl + NaCl
- ◯ $CH_3COOH$ + $CH_3COONa$
- ◯ NaOH + NaCl
- ◯ $HNO_3 + KNO_3$

**Q16** For $Ag_2CrO_4$, $K_{sp}$ = ?
- ◯ $S^2$
- ◯ $4S^3$
- ◯ $27S^4$
- ◯ $2S^2$

**Ruko aur socho**

After the test: in which topics did you get questions wrong? Tick those topics and revise them again.

☐ Topic 1 ☐ Topic 2 ☐ Topic 3 ☐ Topic 4 ☐ Topic 5 ☐ Topic 6 ☐ Topic 7 ☐ Topic 8 ☐ Topic 9



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# Poori book chahiye?

Is sample mein har chapter ke sirf do pages the. Poori **Chemophile Part 1 (for 11th Class)** mein 9 chapters ke **651 pages** hain, har topic isi style mein.

Adeel Sir

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- ✓ Concept pehle, ratna baad mein
- ✓ NCERT ka har question solved
- ✓ Board ke liye English answers
- ✓ Har chapter ke end mein 40 marks ka test

**Teachers aur schools ke liye**

- ✓ NCERT section numbers ke saath
- ✓ Har topic ke ready-made visuals
- ✓ Recap, quick check aur answer key
- ✓ Revision: fact ya formula sheet, top mistakes

## Is sample mein kaun se topics the

| Chapter | Sample topic | Poora chapter |
|---|---|---|
| **1.** Some Basic Concepts | Topic 6: Mole concept | 82 pages |
| **2.** Structure of Atom | Topic 5: Bohr model | 81 pages |
| **3.** Classification of Elements | Topic 6: Ionization enthalpy | 58 pages |
| **4.** Chemical Bonding | Topic 5: VSEPR theory: molecules ki shape | 62 pages |
| **5.** Thermodynamics | Topic 6: Reaction enthalpy aur Hess's law | 68 pages |
| **6.** Equilibrium | Topic 5: Le Chatelier's principle | 73 pages |
| **7.** Redox Reactions | Topic 3: Oxidation number | 52 pages |
| **8.** Organic Chemistry: Basics | Topic 8: Electronic effects aur reaction types | 88 pages |
| **9.** Hydrocarbons | Topic 9: Benzene: structure aur aromaticity | 87 pages |

**Chemophile Part 2 (for 12th Class) bhi ready hai**

10 chapters, 575 pages: usi style mein, usi author ki taraf se.

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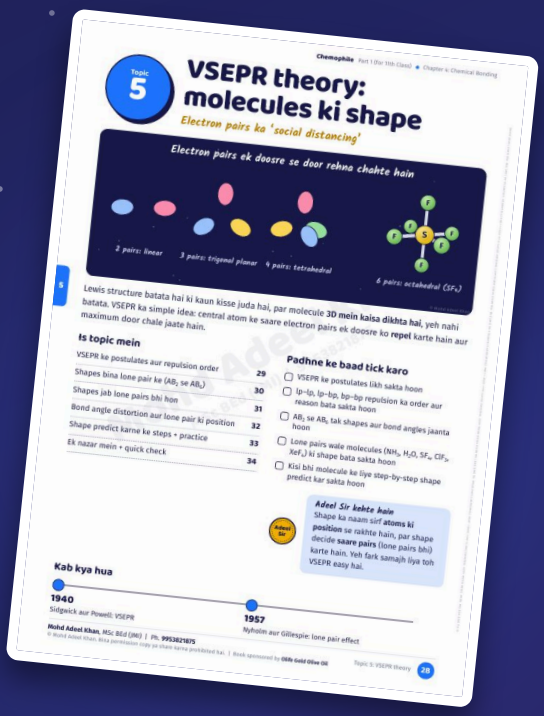
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# Chemophile

**Part 1 (for 11th Class)**

Free sample

*Chemistry ratni nahi, samajhni hai.*




| 9 | 651 | 499 | 1010+ |
|---|---|---|---|
| chapters | colourful pages | figures aur diagrams | practice questions |

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